

Multiple Choice (10 marks)

1. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared
2. A _____ may be a hidden part of a program, a separate infected program a Trojan in disguise of an executable or code in the firmware of any system's hardware.
 - (a) crypter
 - (b) virus
 - (c) backdoor
 - (d) key-logger
3. What are the types of scanning?
 - (a) Port, network, and services
 - (b) Network, vulnerability, and port
 - (c) Passive, active, and interactive
 - (d) Server, client, and network
4. Which of the following does authorization aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
5. A _____ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an array of integers.
 - (a) stack
 - (b) queue
 - (c) external storage
 - (d) buffer

True / False (10 marks)

Answer True or False

6. True or False: Comparing the TTL values of actual and spoofed IP addresses is an effective method for detecting IP address spoofing.
7. True or False: Message nonrepudiation ensures that a sender cannot deny having sent a specific message.
8. True or False: IM Trojans are malicious programs designed to steal logins and passwords for instant messaging applications.
9. True or False: A SYN stealth scan does not complete the TCP handshake, allowing it to remain less detectable.
10. True or False: A session symmetric key can be reused multiple times within different sessions without compromising security.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to sort a given mixed list of integers and strings.
12. Write a function to sum elements in two lists.

End of Paper